

Technology Stack

Date	28 th june 2026
Team ID	xxxxxx
Project Name	Rising waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

c	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Provides a simple, responsive, and interactive user interface for entering environmental parameters and displaying flood prediction results.

2	Backend / Server-Side	Python, Flask	Processes user requests, loads the trained ML model, performs predictions, and returns results to the frontend.
3	Database / Data Storage	CSV Dataset (Flood_prediction.csv)	Stores historical flood-related environmental data used for training and testing the machine learning model.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Flask Local Server (or Render, if deployed)	Used to run and test the web application locally. Render can be used for cloud deployment and public access.
5	Version Control & CI/CD	Git & GitHub	Git helps track code changes, while GitHub stores the project repository for collaboration and version management.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio,	Joblib, Visual Studio Code	Joblib is used to save and load the trained ML model efficiently. Visual Studio Code is used for project development and debugging.

